

Roselle (Hibiscus sabdariffa L.) and soybean oil effects on quality characteristics of pork patties studied by response surface methodology.

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Response surface methodology was used to investigate the effect and interactions of processing variables such as roselle extract (0.1-1.3%), soybean oil (5-20%) on physicochemical, textural and sensory properties of cooked pork patties. It was found that reduction in thickness, pH, L* and b* values decreased; however, water-holding capacity, reduction in diameter and a* values increased, respectively, as the amount of roselle increased. Soybean oil addition increased water-holding capacity, reduction in thickness, b* values of the patties. The hardness depended on the roselle and soybean oil added, as its linear effect was negative at $p < 0.01$. The preference of color, tenderness, juiciness, and overall quality depend on the addition of roselle and soybean oil. The maximum overall quality score (5.42) was observed when 12.5 g of soybean oil and 0.7 g of roselle extract was added. The results of this optimization study would be useful for meat industry that tends to increase the product yield for patties using the optimum levels of ingredients by RSM. Copyright © 2013 Elsevier Ltd. All rights reserved.